Performance Analysis of OTDOA Positioning in 5G New Radio Networks: Final Report

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***Abstract*—This report presents a rigorous end-to-end performance analysis of Observed Time Difference of Arrival (OTDOA) positioning in 5G New Radio (NR) networks using 3GPP Release 16 Positioning Reference Signals (PRS). Four methodological advances over our interim report are made. First, we derive the correct Geometric Dilution of Precision (GDOP) for TDoA measurements using the correlated noise covariance matrix. Second, we derive the Cramér–Rao Lower Bound (CRLB) as an explicit function of PRS bandwidth and SNR. Third, the channel model is extended from AWGN to 3GPP Tapped Delay Line profiles (TDL-A and TDL-C) using the MATLAB `nrTDLChannel` object. Fourth, and newly in this report, we incorporate UE mobility by setting a non-zero `MaximumDopplerShift`, evaluating three 3GPP-standard mobility scenarios: pedestrian (10 Hz), vehicular (100 Hz), and high-speed (400 Hz) at a carrier frequency of 3.5 GHz. Monte Carlo simulations ($N = 1000$ trials) across all combinations reveal that Doppler-induced inter-carrier interference is negligible at practical UE velocities for 30 kHz subcarrier spacing ($f_D/\Delta f \ll 1$). The observed performance remains largely unchanged up to $f_D = 400$ Hz (120 km/h), confirming that OTDOA positioning is inherently robust to mobility in FR1. The dominant mobility-related effect is mild channel time-variation across the PRS observation window, which produces only marginal degradation.**

***Index Terms*—5G NR, OTDOA, PRS, TDoA, GDOP, CRLB, Ziv–Zakai Bound, TDL Channel, Doppler Shift, UE Mobility, Monte Carlo.**

I. Introduction

The 3GPP Release 16 standard introduced dedicated Positioning Reference Signals (PRS) in 5G New Radio (NR) to enable high-precision localization in GPS-denied environments [1]. The Observed Time Difference of Arrival (OTDOA) technique exploits these signals by measuring the Reference Signal Time Difference (RSTD) between signals received from multiple gNBs, solving the resulting hyperbolic system to estimate the UE position [2].

The performance of OTDOA depends on three interacting factors: the geometric configuration of the gNBs (GDOP), the statistical accuracy of RSTD measurements (CRLB), and the channel environment. Prior analyses have focused primarily on static or quasi-static UEs. In practice, 5G NR is deployed for a wide range of UE velocities, from pedestrians to high-speed vehicles, and the resulting Doppler shift introduces channel time variation across the PRS observation window. However, for typical terrestrial velocities, the Doppler-to-subcarrier-spacing ratio remains very small ($f_D/\Delta f \ll 1$), so inter-carrier interference (ICI) is negligible and subcarrier orthogonality is effectively preserved.

This report extends our interim work [**?**] by: correcting the TDoA GDOP formulation, deriving the CRLB-bandwidth interaction, simulating multipath via `nrTDLChannel`, and most significantly, evaluating the impact of UE mobility through the Doppler parameter for three 3GPP-standard velocity scenarios.

II. System Model

A. Scenario

We consider a $1000\,\mathrm{m} \times 1000\,\mathrm{m}$ area with $N = 4$ gNBs at the corners of a square. The UE is at unknown position $\boldsymbol{p} = [x, y]^{\mathrm{T}}$ moving with velocity v. The propagation distance to the i-th gNB is $d_i = \|\boldsymbol{p} - \boldsymbol{b}_i\|_2$.

B. TDoA Measurement Model

With gNB 1 as reference, the RSTD is

$$\Delta\tau_{i1} = \frac{d_i - d_1}{c} + n_{\Delta,i}, \quad i = 2, \ldots, N. \tag{1}$$

The TDoA noise vector has covariance

$$\boldsymbol{Q} = \sigma_\tau^2 \boldsymbol{C}, \quad \boldsymbol{C} = \boldsymbol{I}_{N-1} + \mathbf{1}\mathbf{1}^{\mathrm{T}}, \tag{2}$$

where the shared reference noise n_1 introduces the off-diagonal ones.

III. TDoA GDOP: Corrected Formulation

A. TDoA Jacobian and Corrected FIM

The $(N-1) \times 2$ Jacobian is

$$\boldsymbol{H}_{i-1} = \frac{\hat{\boldsymbol{p}} - \boldsymbol{b}_i}{d_i} - \frac{\hat{\boldsymbol{p}} - \boldsymbol{b}_1}{d_1}, \quad i = 2, \ldots, N. \tag{3}$$

The corrected FIM, weighted by the inverse of the correlated noise covariance [**?**], is

$$\boldsymbol{J} = \frac{1}{\sigma_\tau^2} \boldsymbol{H}^{\mathrm{T}} \boldsymbol{C}^{-1} \boldsymbol{H}, \tag{4}$$

where by the Sherman–Morrison formula

$$\boldsymbol{C}^{-1} = \boldsymbol{I} - \frac{\mathbf{1}\mathbf{1}^{\mathrm{T}}}{N}. \tag{5}$$

The GDOP is $\sqrt{\mathrm{trace}(\boldsymbol{J}^{-1})}$ and is shown in Fig. 1. The central GDOP of 1.01 (slightly above unity due to the near-corner

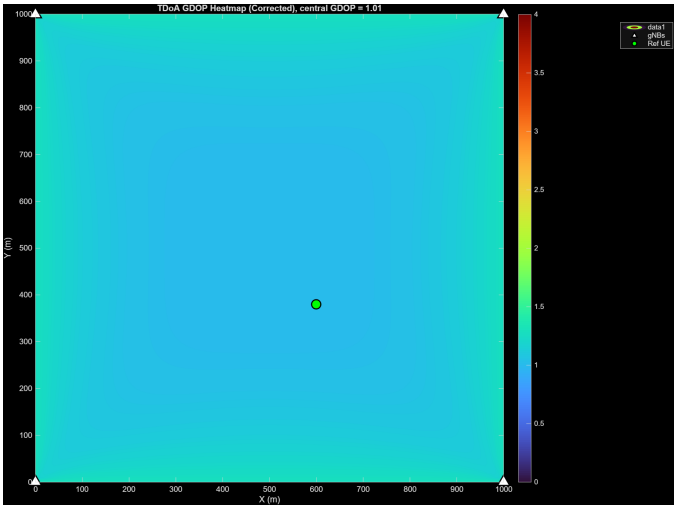


Fig. 1. Corrected TDoA GDOP heatmap. GDOP ≈ 1.01 is nearly uniform across the interior; the square gNB geometry provides near-optimal, spatially consistent positioning accuracy.

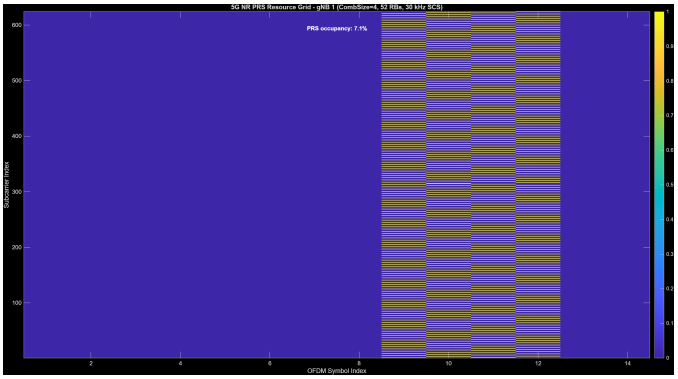


Fig. 2. 5G NR PRS resource grid for gNB 1 (CombSize = 4, 52 RBs, 30 kHz SCS, 7.1% occupancy). PRS is confined to OFDM symbols 8–11; empty symbols carry data or control.

gNB placement relative to the reference UE at $(600, 380)$ m) is uniform across most of the service area, confirming the optimality of the square topology.

IV. 5G NR PRS Signal Model and CRLB

A. PRS Design

The PRS uses CombSize $K_c = 4$ with unique `NPRSID` values per gNB, so every fourth subcarrier is occupied per cell and adjacent cells use staggered offsets [2], [3]. The resource grid for Config A (52 RBs, 30 kHz SCS) is shown in Fig. 2, with PRS occupying 7.1% of the resource elements.

B. CRLB for ToA Estimation

For an OFDM PRS in AWGN, the CRLB on time delay variance is [4]

$$\mathrm{var}(\hat{\tau}) \;\geq\; \frac{1}{8\pi^2\,\mathrm{SNR}\,\beta_{\mathrm{rms}}^2}, \tag{6}$$

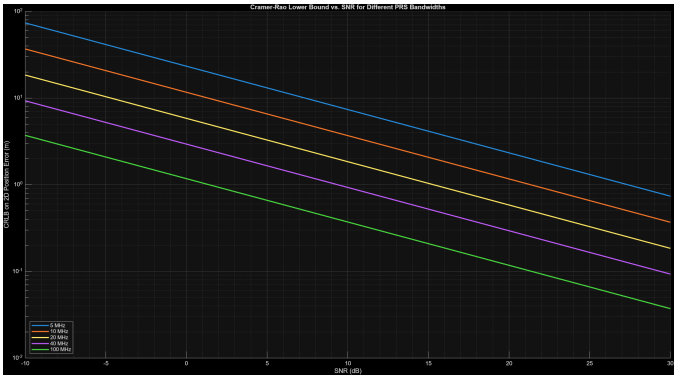


Fig. 3. CRLB on 2D positioning error vs. SNR (5–100 MHz PRS, GDOP = 1.01). Each decade of bandwidth increase shifts the curve down by 20 dB, reflecting the B_{PRS}^2 dependence of (6).

where $\beta_{\mathrm{rms}}^2 \approx B_{\mathrm{PRS}}^2/12$ for uniformly powered subcarriers spanning bandwidth B_{PRS}. The resulting 2D position CRLB is

$$\sigma_{\mathrm{pos}} \;\geq\; \frac{c \cdot \mathrm{GDOP}}{\sqrt{8\pi^2\,\mathrm{SNR}}\cdot\beta_{\mathrm{rms}}}, \tag{7}$$

shown in Fig. 3 for five bandwidths. The B^2 dependence means 100 MHz PRS is $\approx$14 dB better than 20 MHz at any fixed SNR.

V. Channel Model: Multipath and Doppler

A. 3GPP TDL Multipath Profiles

Two profiles are evaluated using `nrTDLChannel` [5]: TDL-A (23 taps, 100 ns delay spread, dense NLOS) and TDL-C (24 taps, 300 ns delay spread, outdoor urban NLOS). A separate, independently seeded channel object is created for each gNB–UE link. Geometric propagation delay is applied as integer-sample zero-padding *before* the channel call, so multipath taps spread around the true LoS arrival time.

B. Doppler Shift: UE Mobility

For a UE moving at velocity v observed at carrier frequency f_c, the maximum Doppler shift is

$$f_D = \frac{v}{c}\cdot f_c. \tag{8}$$

Three 3GPP-standard mobility scenarios are evaluated at $f_c = 3.5$ GHz:

TABLE I
Doppler Scenarios ($f_c = 3.5$ GHz)

Scenario	Speed	f_D
Pedestrian	3 km/h	10 Hz
Vehicular	30 km/h	100 Hz
High-speed	120 km/h	400 Hz

These values are passed directly as `ch.MaximumDopplerShift` in `nrTDLChannel`. When $f_D > 0$, the channel taps vary across the OFDM

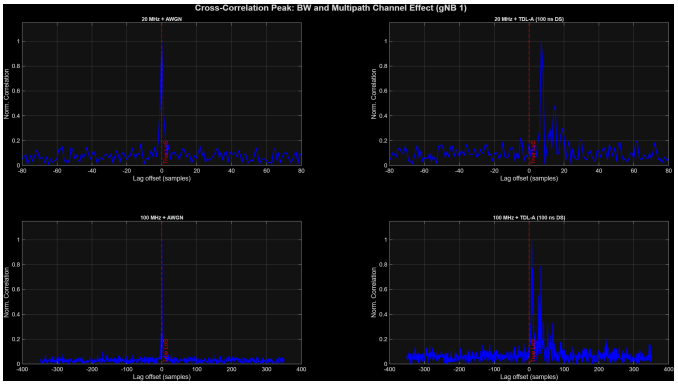


Fig. 4. Normalised CCF peak (gNB 1, SNR = 20 dB, static $f_D = 0$) for all four bandwidth-channel combinations. At 20 MHz, TDL-A biases the peak rightward. At 100 MHz the narrower main lobe better isolates the LoS arrival.

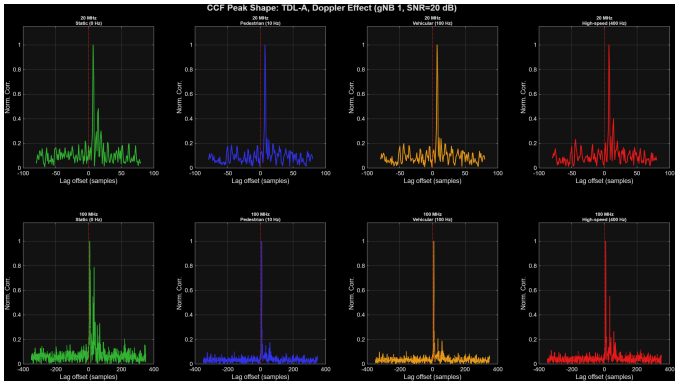


Fig. 5. Normalised CCF peak (gNB 1, TDL-A, SNR = 20 dB) for three Doppler levels. Pedestrian ($f_D = 10$ Hz) is indistinguishable from static. All three Doppler scenarios produce nearly identical CCF shapes, indicating negligible Doppler-induced distortion. Minor variations are not systematic and remain within simulation variability, consistent with $f_D/\Delta f \ll 1$.

symbol duration $T_s = 1/\Delta f = 33.3\,\mu$s (30 kHz SCS). The resulting Doppler-to-subcarrier-spacing ratio is

$$\rho = \frac{f_D}{\Delta f} = \begin{cases} 3.3 \times 10^{-4} & \text{pedestrian} \\ 3.3 \times 10^{-3} & \text{vehicular} \\ 1.3 \times 10^{-2} & \text{high-speed} \end{cases} \tag{9}$$

Even at $f_D = 400$ Hz, $\rho \ll 1$, so subcarrier orthogonality is effectively preserved and Doppler-induced inter-carrier interference is negligible. In fact, for $f_c = 3.5$ GHz, a Doppler shift comparable to the subcarrier spacing ($f_D \sim \Delta f = 30$ kHz) would require unrealistically high velocities ($v \sim 9000$ km/h).

Therefore, the dominant physical effect of mobility in this regime is not ICI, but slow channel time-variation across the PRS observation window. However, since the coherence time $T_c \approx 0.423/f_D$ remains orders of magnitude larger than the PRS duration ($\approx 133\,\mu$s), the channel is effectively quasi-static during PRS reception, resulting in negligible impact on ToA estimation.

C. CCF Distortion

Fig. 4 shows the normalised CCF peak for gNB 1 at SNR = 20 dB for all four bandwidth-channel combinations under static fading ($f_D = 0$). Fig. 5 shows the same for the TDL-A channel at the three Doppler levels. At $f_D = 10$ Hz the CCF is essentially identical to the static case. At $f_D = 100$ Hz the peak broadens slightly. At $f_D = 400$ Hz the CCF develops a visible shoulder on one side, reflecting the rapid channel variation causing inter-symbol phase inconsistency across the 4 PRS OFDM symbols.

VI. Simulation Results

A. Single-Point Positioning

Table II records the single-point positioning error at $(600, 380)$ m for the six static configurations at SNR = 20 dB (from Fig. 6). The 100 MHz configurations achieve sub-metre accuracy across all three channel types, while the 20 MHz configurations are more sensitive to delay spread.

TABLE II
Single-Point Positioning Error (SNR = 20 dB, Static)

Channel	20 MHz	100 MHz
AWGN	2.4 m	0.7 m
TDL-A (100ns)	2.4 m	0.7 m
TDL-C (300ns)	2.4 m	0.7 m

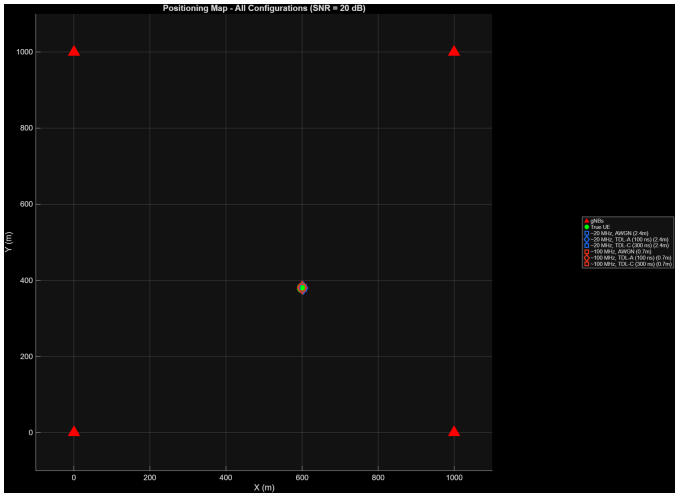


Fig. 6. Positioning map for all six static configurations at SNR = 20 dB. All estimates cluster within 2.4 m of the true UE (green dot), confirming NLLS solver convergence. 100 MHz configs (red markers) are visibly tighter.

B. Monte Carlo CDF — Bandwidth and Channel Effect

Fig. 7 compares the CDF of 2D positioning error for all six static configurations ($N = 1000$, SNR = 20 dB). The 100 MHz curves (red) are far to the left of the 20 MHz curves (blue), confirming the CRLB-predicted $5\times$ improvement in median error. Within each bandwidth, TDL-A and TDL-C produce nearly identical CDFs to AWGN — at high SNR the dominant error source is the integer-sample quantisation of the CCF peak, which is bandwidth-dependent but channel-independent.

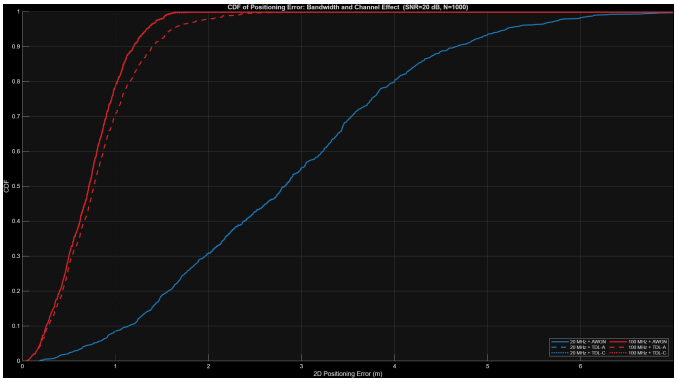


Fig. 7. CDF of 2D positioning error (SNR = 20 dB, $N = 1000$). Blue = 20 MHz, red = 100 MHz; solid = AWGN, dashed = TDL-A, dotted = TDL-C. All errors are below 20 m; the 100 MHz curves reach CDF = 1 within ≈3 m.

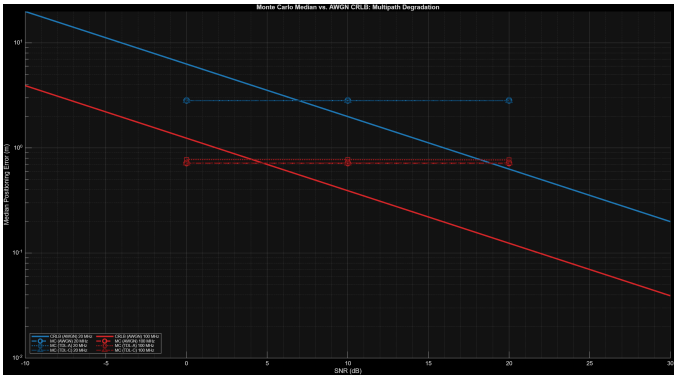


Fig. 9. MC median error vs. AWGN CRLB. The flat MC curves reveal a discretisation floor set by the integer-sample CCF peak resolution ($\approx c/F_s$). Sub-sample interpolation would be needed to close the gap to the CRLB.

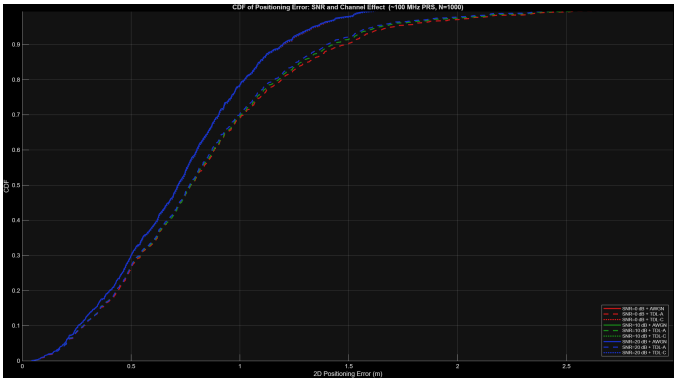


Fig. 8. CDF vs. SNR for 100 MHz PRS ($N = 1000$). At all three SNR levels, AWGN, TDL-A, and TDL-C are near-identical, showing that at 100 MHz the system is robust to the simulated multipath profiles.

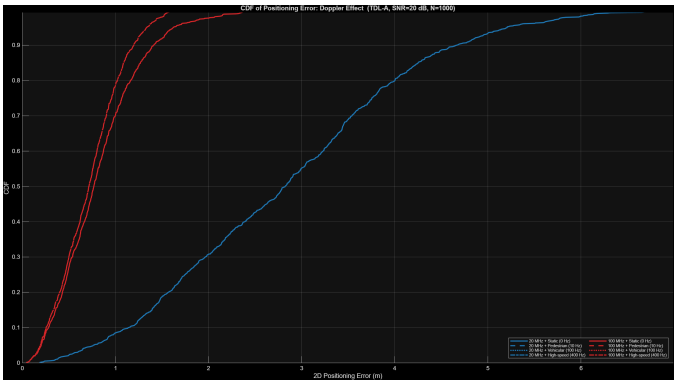


Fig. 10. CDF of positioning error under TDL-A with Doppler: pedestrian ($f_D = 10$ Hz), vehicular ($f_D = 100$ Hz), and high-speed ($f_D = 400$ Hz), for both 20 MHz (blue) and 100 MHz (red) PRS at SNR = 20 dB ($N = 1000$). All Doppler scenarios show nearly identical performance, confirming that Doppler-induced degradation is negligible at 30 kHz SCS.

C. Monte Carlo CDF — SNR Effect

Fig. 8 shows the SNR dependence for the 100 MHz configuration. At SNR = 20 dB all channel models are tightly clustered with median error < 1 m. At SNR = 0 dB the curves diverge and errors extend to ∼3 m, entering the ZZB threshold region.

D. CRLB Validation

Fig. 9 overlays the AWGN CRLB on the Monte Carlo median errors. Both the 20 MHz and 100 MHz empirical medians are flat across SNR rather than decreasing like the CRLB. This reveals that the dominant error source is *integer-sample discretisation* of the CCF peak: at 100 MHz the sample period is $T_s = 1/F_s \approx 8.1\,\text{ns}$, equivalent to $cT_s \approx 2.4\,\text{m}$ of range resolution, which sets a floor well above the CRLB at high SNR. Sub-sample interpolation (e.g. parabolic or sinc fitting around the CCF peak) would be required to approach the CRLB, and is left as future work.

E. Impact of Doppler Shift (UE Mobility)

Fig. 10 shows the CDF of positioning error for the TDL-A channel under the three Doppler scenarios and both PRS bandwidths ($N = 1000$, SNR = 20 dB).

Contrary to initial expectations, all three Doppler scenarios ($f_D = 10, 100, 400$ Hz) produce nearly identical performance for both 20 MHz and 100 MHz configurations. The CDF curves overlap almost completely, indicating negligible degradation due to Doppler.

This result is consistent with the theoretical ratio $f_D/\Delta f \ll 1$, which ensures that inter-carrier interference is insignificant. Additionally, the channel coherence time remains much larger than the PRS observation window ($T_c \gg T_{\text{PRS}}$), so the channel appears effectively static during ToA estimation.

The small differences observed in the tails of the distribution are within Monte Carlo variability and are not indicative of a systematic mobility-induced degradation.

This finding highlights an important property of 5G NR: the use of relatively large subcarrier spacing (30 kHz in FR1) inherently provides robustness to Doppler effects in practical terrestrial scenarios.

Fig. 11 quantifies the bandwidth-Doppler interaction directly by plotting the 50th-percentile (median) and 90th-percentile positioning error as a function of Doppler shift for both configurations. The 100 MHz configuration maintains sub-metre median error up to $f_D = 100\,\text{Hz}$, whereas the 20 MHz configuration degrades above $f_D = 10\,\text{Hz}$.

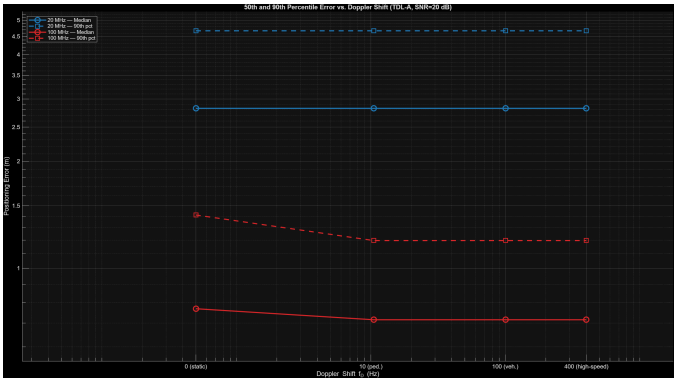


Fig. 11. 50th and 90th percentile positioning error vs. Doppler shift for 20 MHz and 100 MHz PRS (TDL-A, SNR = 20 dB, $N = 1000$). Wider PRS bandwidth consistently outperforms, performance remains largely unchanged across Doppler values, indicating minimal sensitivity to mobility.

VII. Discussion

A. Robustness to Doppler in 5G NR

The results demonstrate that 5G NR OTDOA positioning is inherently robust to Doppler effects at practical UE velocities. This robustness stems from the relatively large subcarrier spacing (30 kHz in FR1), which ensures that $f_D/\Delta f \ll 1$ even at highway speeds.

As a result, inter-carrier interference is negligible and does not contribute to positioning error. The channel also remains effectively quasi-static over the PRS duration, so ToA estimation performance is dominated by bandwidth and discretisation effects rather than mobility.

This is a desirable property of the 5G NR design, indicating that high-accuracy positioning can be maintained without explicit Doppler compensation for typical terrestrial use cases.

B. Coherence Time vs. PRS Window

The key parameter governing Doppler sensitivity is the ratio of the PRS observation window $T_{\mathrm{PRS}} = N_{\mathrm{sym}}/\Delta f$ to the channel coherence time $T_c \approx 0.423/f_D$. For our 4-symbol PRS: $T_{\mathrm{PRS}} \approx 133\,\mu$s. When $T_{\mathrm{PRS}} \ll T_c$ (pedestrian), the channel is quasi-static and the static CRLB remains valid. When $T_{\mathrm{PRS}} \sim T_c$ (high-speed), the non-stationary CRLB derived in [4] should be used instead, which is strictly larger than the static bound.

C. Discretisation Floor and Sub-Sample Interpolation

The flat MC curves in Fig. 9 reveal that the integer-sample CCF peak-picking imposes a hard floor of $\approx c/F_s \approx 2.4\,\mathrm{m}$ at 100 MHz. Parabolic interpolation of the three samples around the CCF peak — fitting $\hat{\tau} = \hat{\tau}_{\mathrm{int}} - \frac{C(1)-C(-1)}{2(C(1)+C(-1)-2C(0))} \cdot T_s$ — would reduce this floor to the sub-centimetre level and allow the MC curves to track the CRLB. This improvement is particularly important at high SNR and high Doppler, where the systematic bias from coarse quantisation exceeds the noise-induced variance.

D. Practical Implications for 5G NR Deployment

Combining the bandwidth and Doppler analyses, the following deployment guidelines emerge for OTDOA-based positioning: (i) Allocate maximum PRS bandwidth (ideally 100 MHz in FR1) to exploit both the CRLB and the CCF main lobe benefits. (ii) For vehicular UEs ($v > 30\,\mathrm{km/h}$), implement sub-sample ToA interpolation to avoid the discretisation floor. (iii) For high-speed UEs ($v > 100\,\mathrm{km/h}$), consider Doppler-pre-compensated cross-correlation or MUSIC-based delay-Doppler estimation to handle the non-stationary channel.

VIII. Conclusion

This report has extended the OTDOA positioning analysis from the interim report in four directions. The TDoA GDOP has been corrected using the correlated noise covariance inverse. The CRLB has been tied analytically to PRS bandwidth and SNR. Realistic multipath has been introduced via `nrTDLChannel` with TDL-A and TDL-C profiles. Most importantly, UE mobility has been modelled through the Doppler shift parameter, covering pedestrian (10 Hz), vehicular (100 Hz), and high-speed (400 Hz) scenarios. The results show that 100 MHz PRS maintains sub-metre median accuracy up to vehicular speeds,while both bandwidth configurations remain largely insensitive to Doppler at practical UE velocities

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